eprint 2026/723

Hint-Free Multi-Signatures

Dennis Hofheinz
ETH Zürich

Michael Reichle
INSAIT

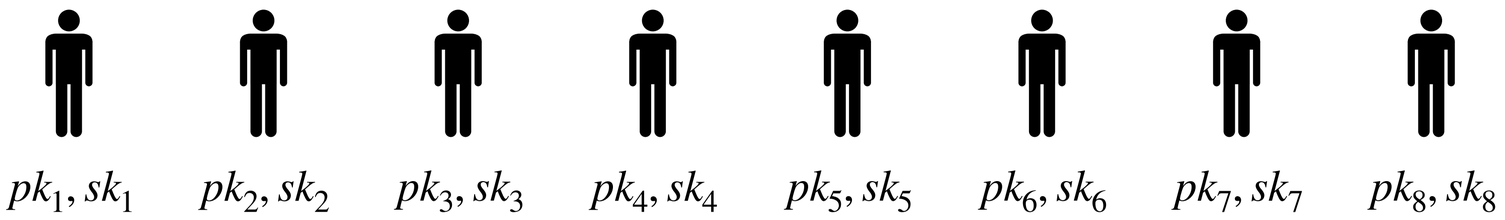
Benedikt Wagner
Ethereum Foundation

Non-Interactive Multi-Signatures

Aggregating Signatures in an unreliable network

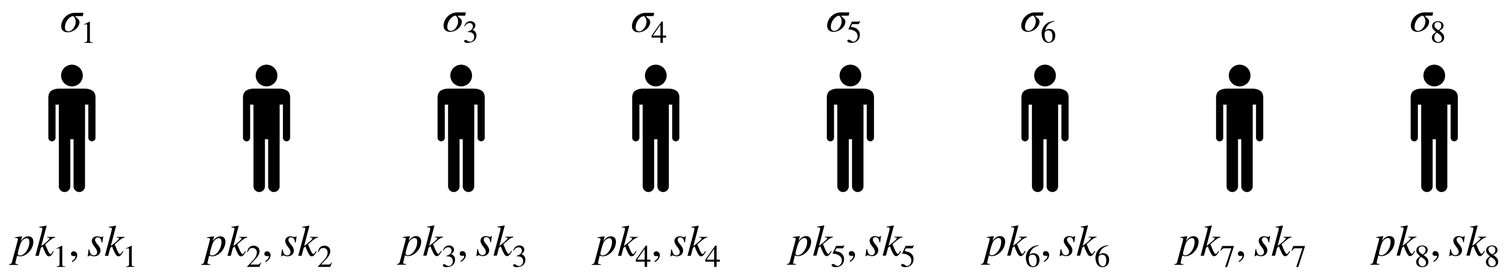
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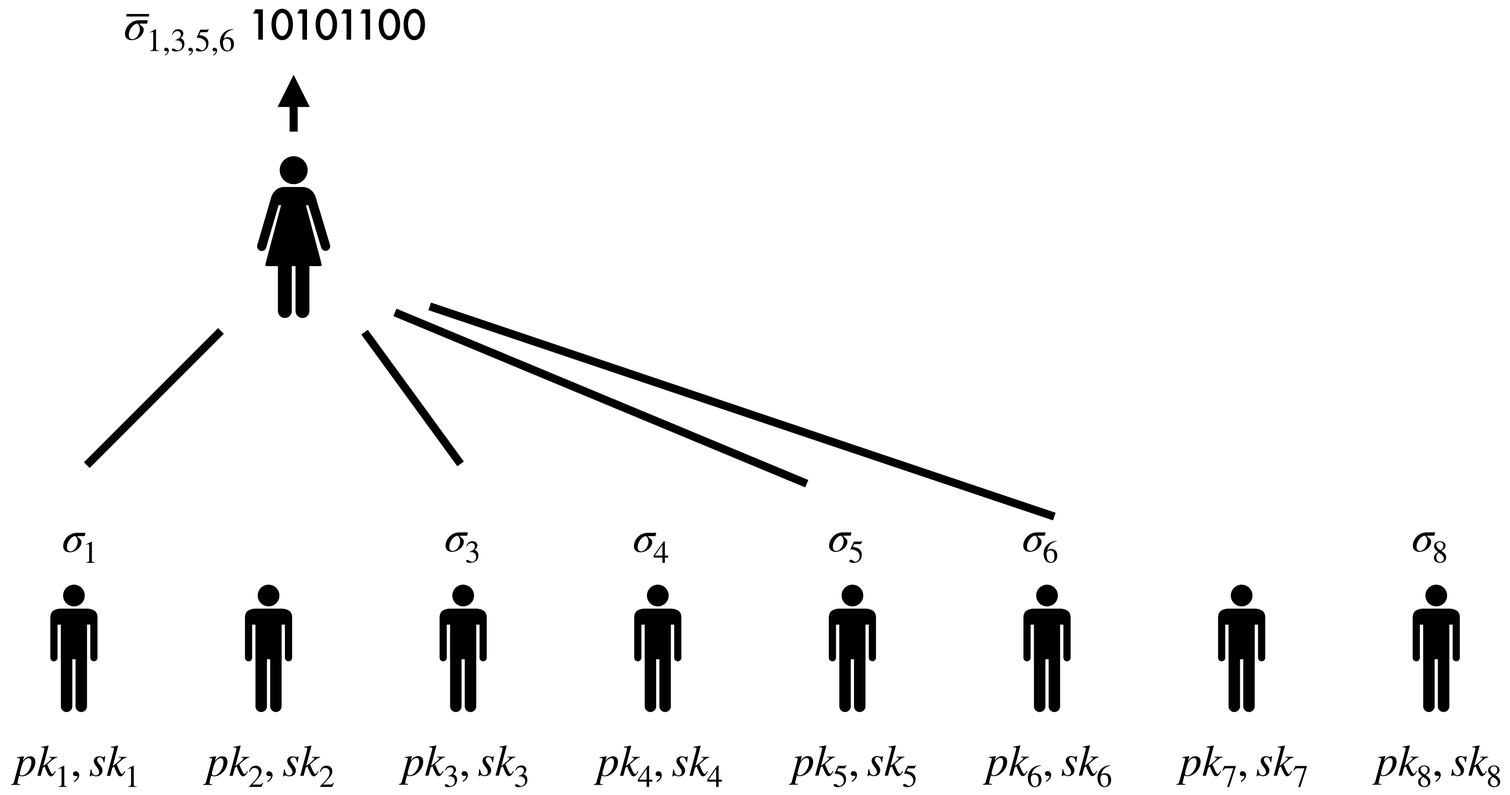
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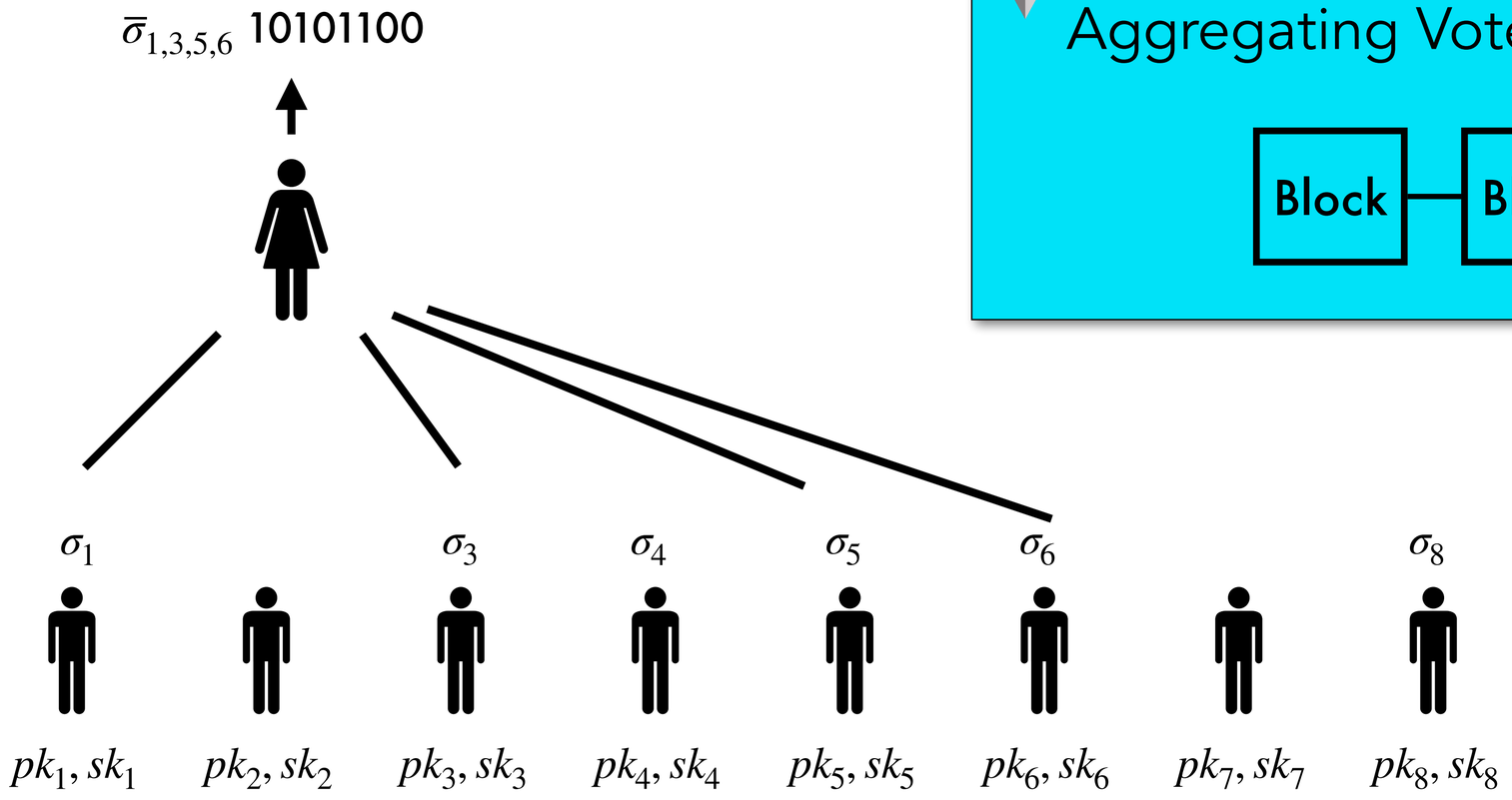
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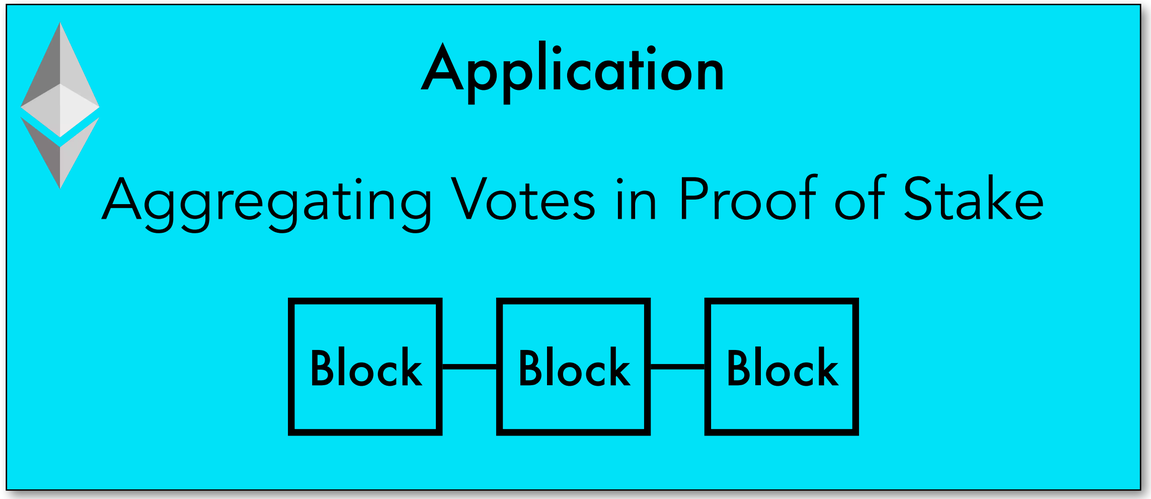
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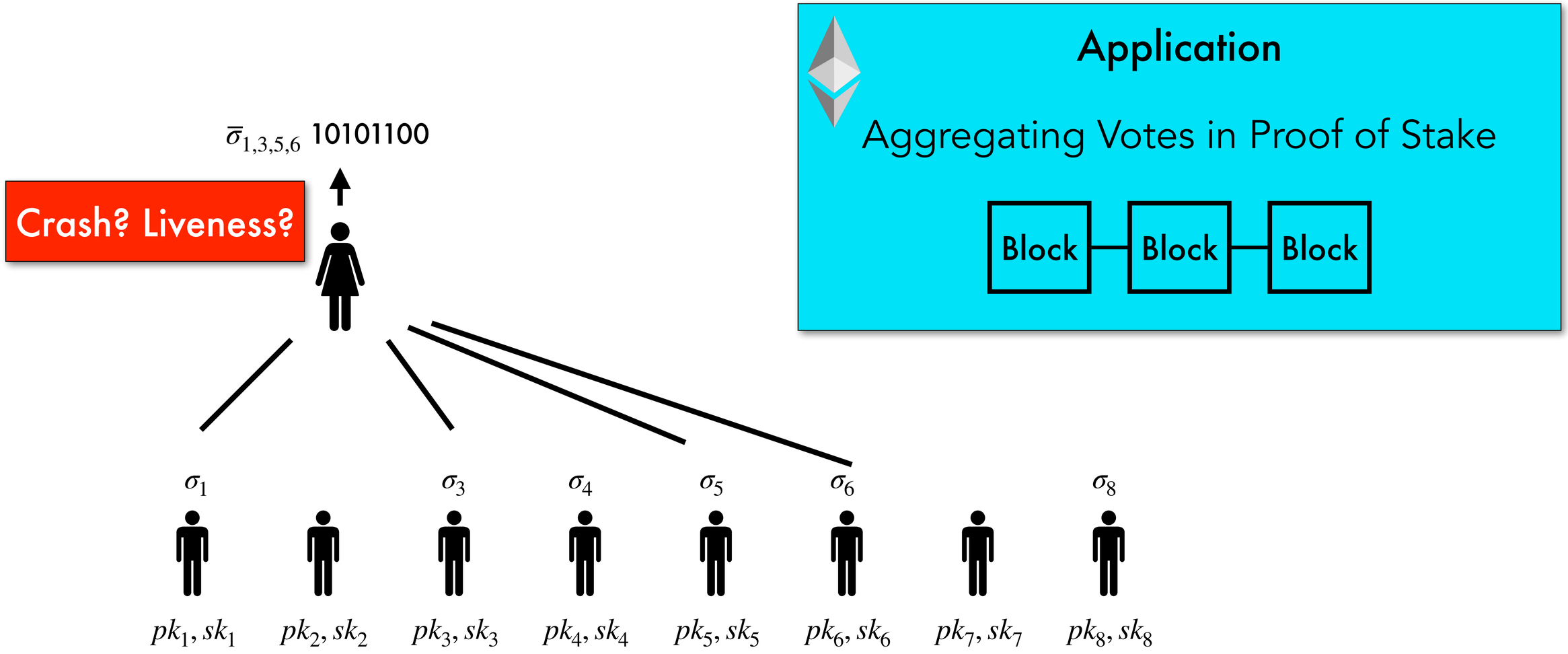
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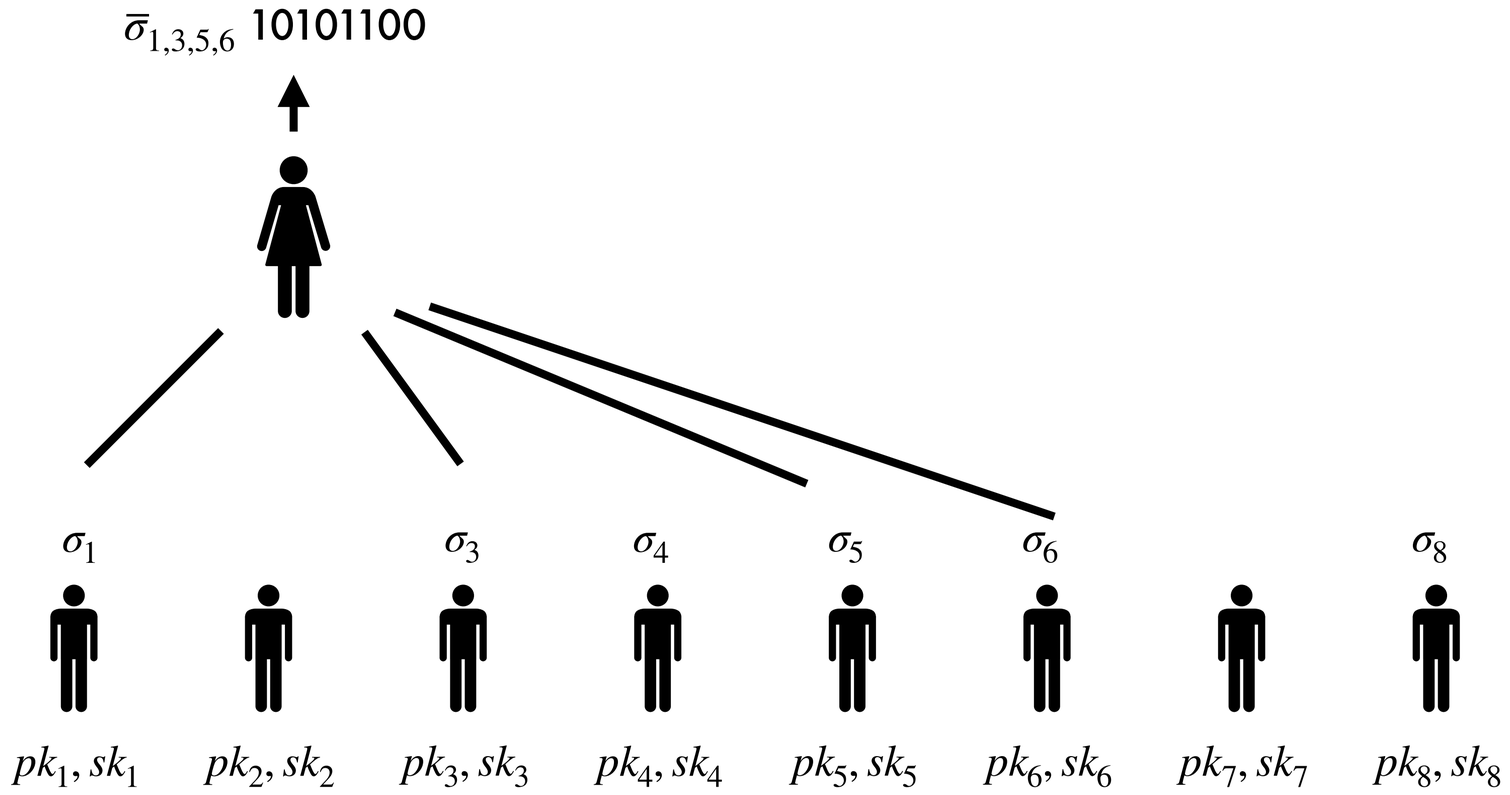
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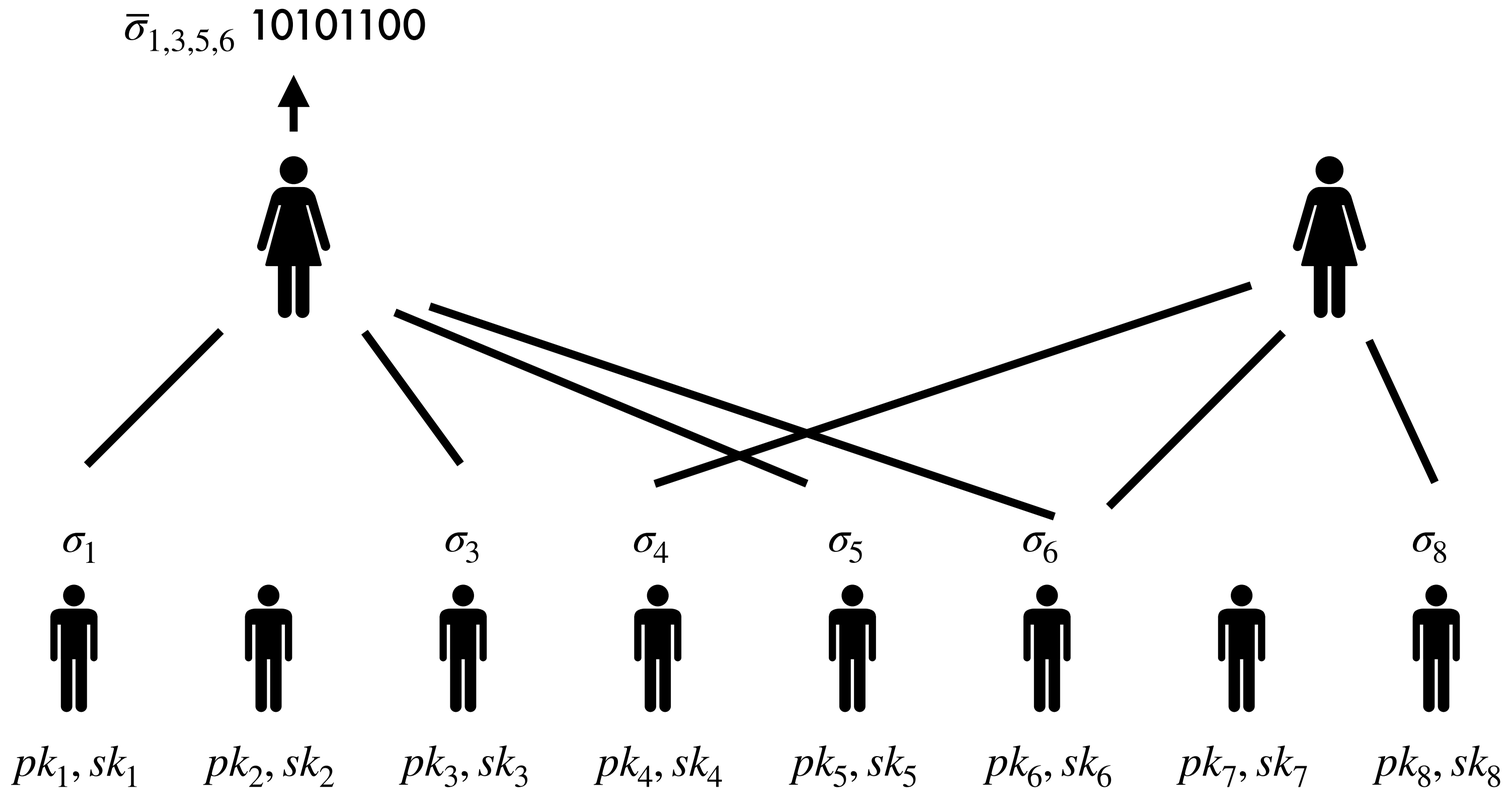
Non-Interactive Multi-Signatures

Redundancy for Liveness



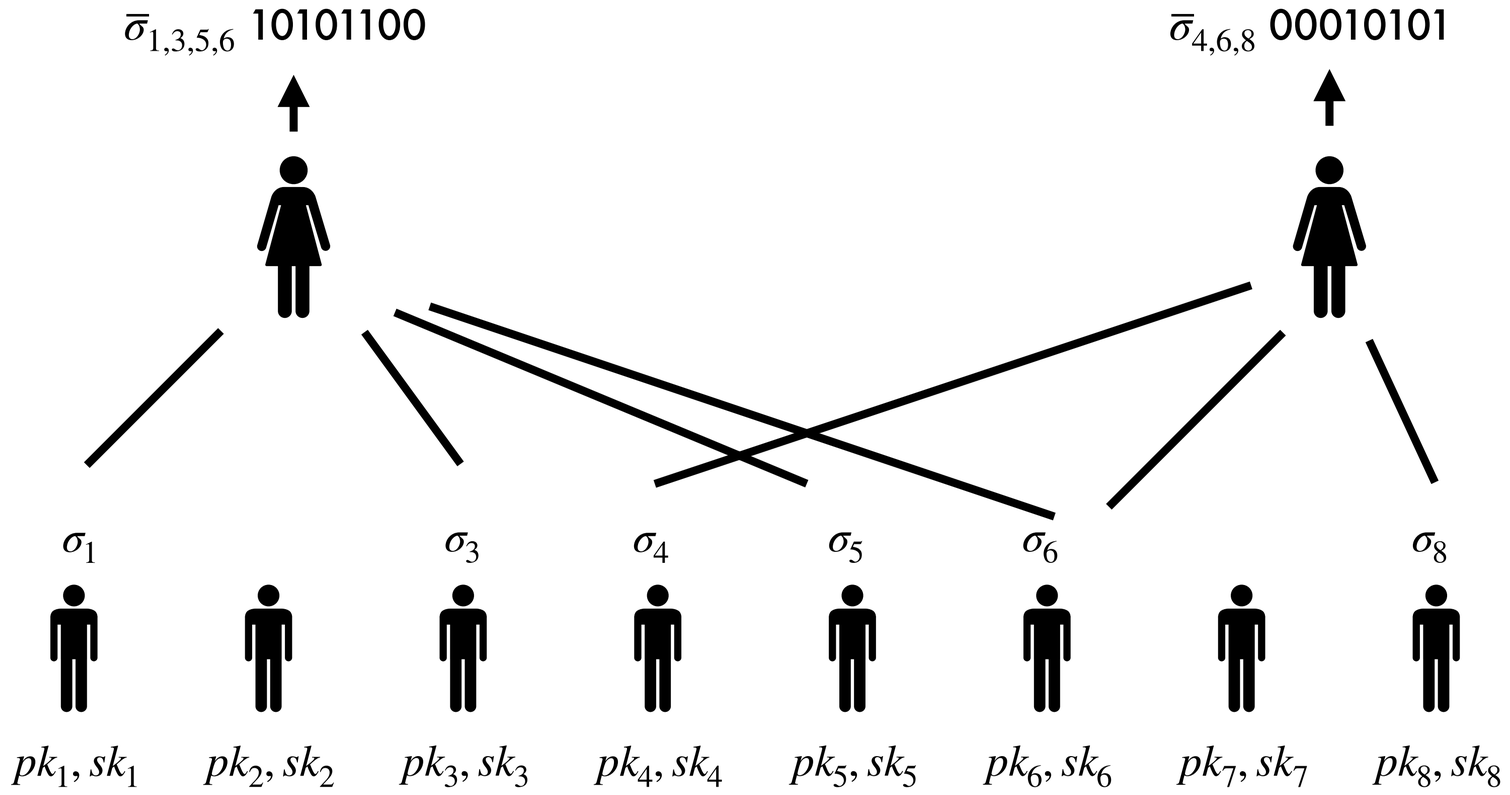
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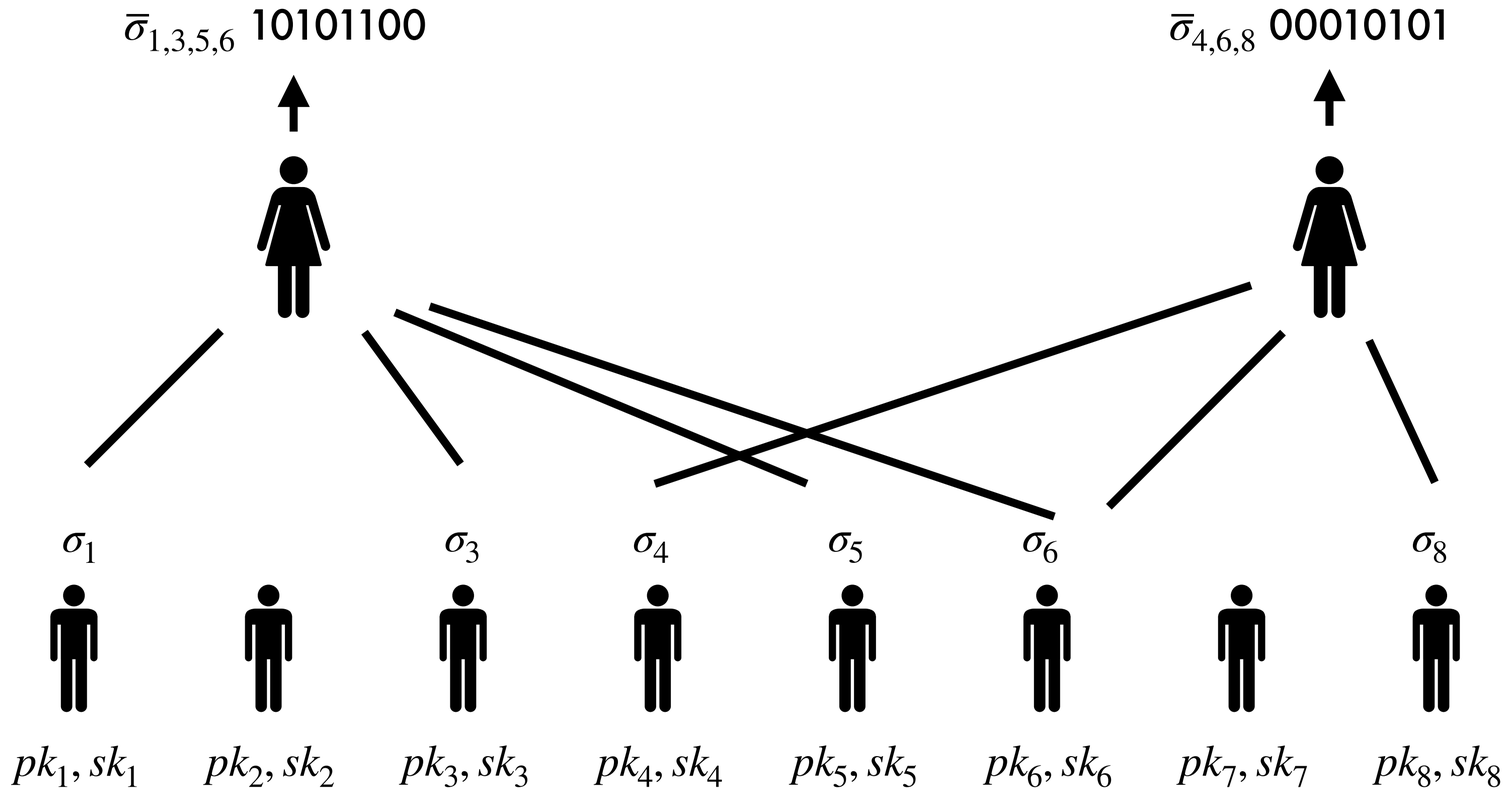
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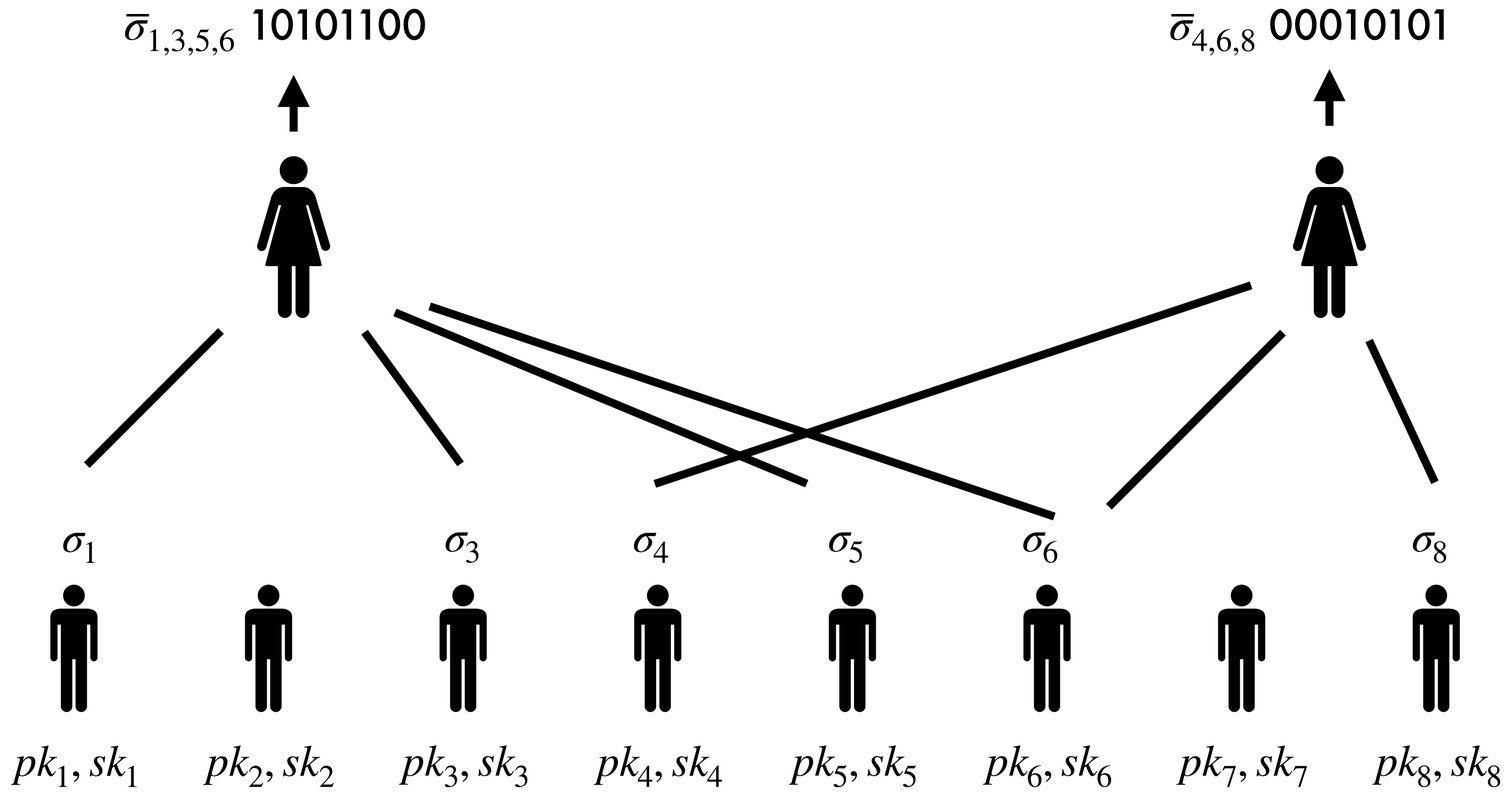
Redundancy for Liveness



One sig + one bit vector per aggregator

Non-Interactive Multi-Signatures

Redundancy for Liveness



One sig + one bit vector per aggregator

Can we aggregate aggregates?

BLS Signatures and Multi-Signatures

Prime Order Pairing Groups

Secret Key

Public Key

Signing

Verification

Aggregation

BLS Signatures and Multi-Signatures

Prime Order Pairing Groups $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$ $|\mathbb{G}| = p$

Secret Key $sk \leftarrow \mathbb{Z}_p$

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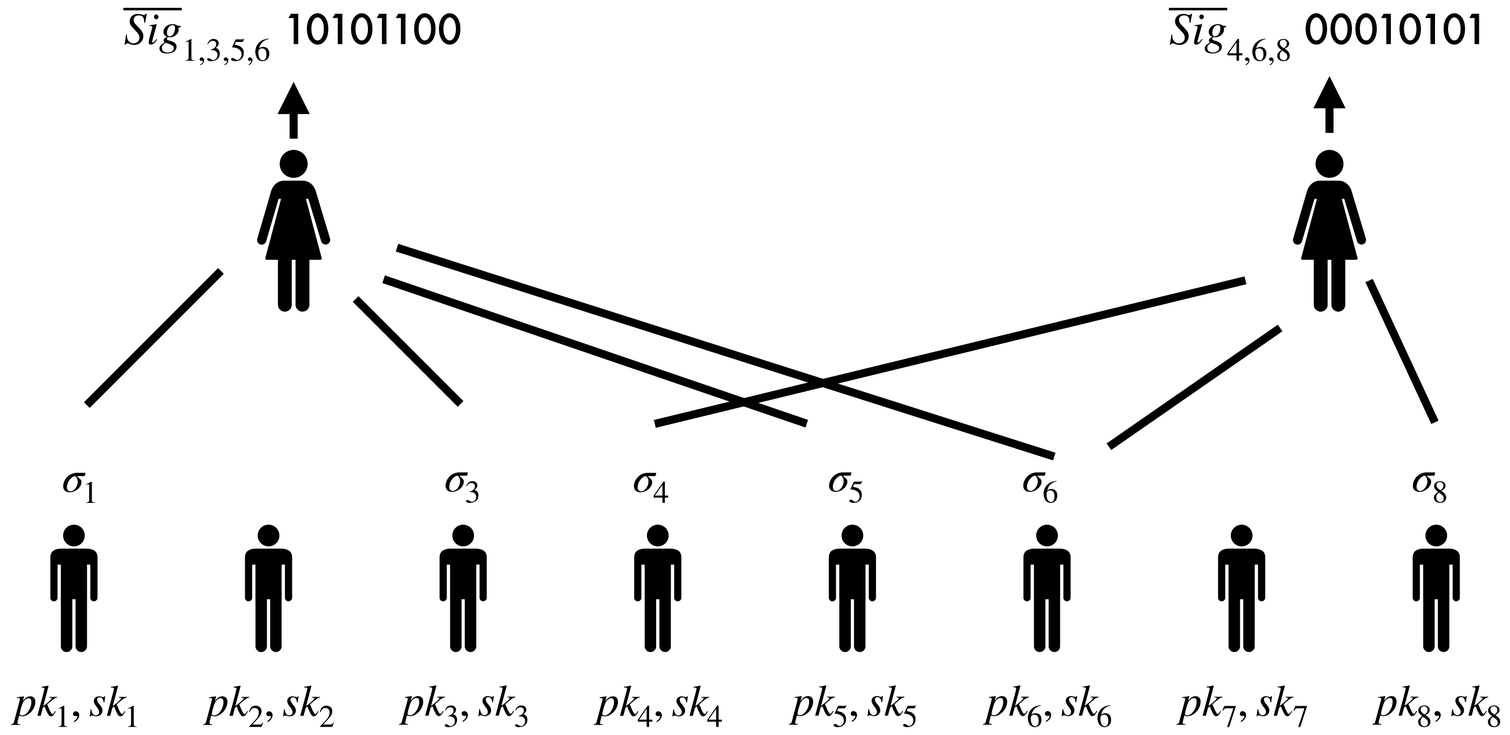
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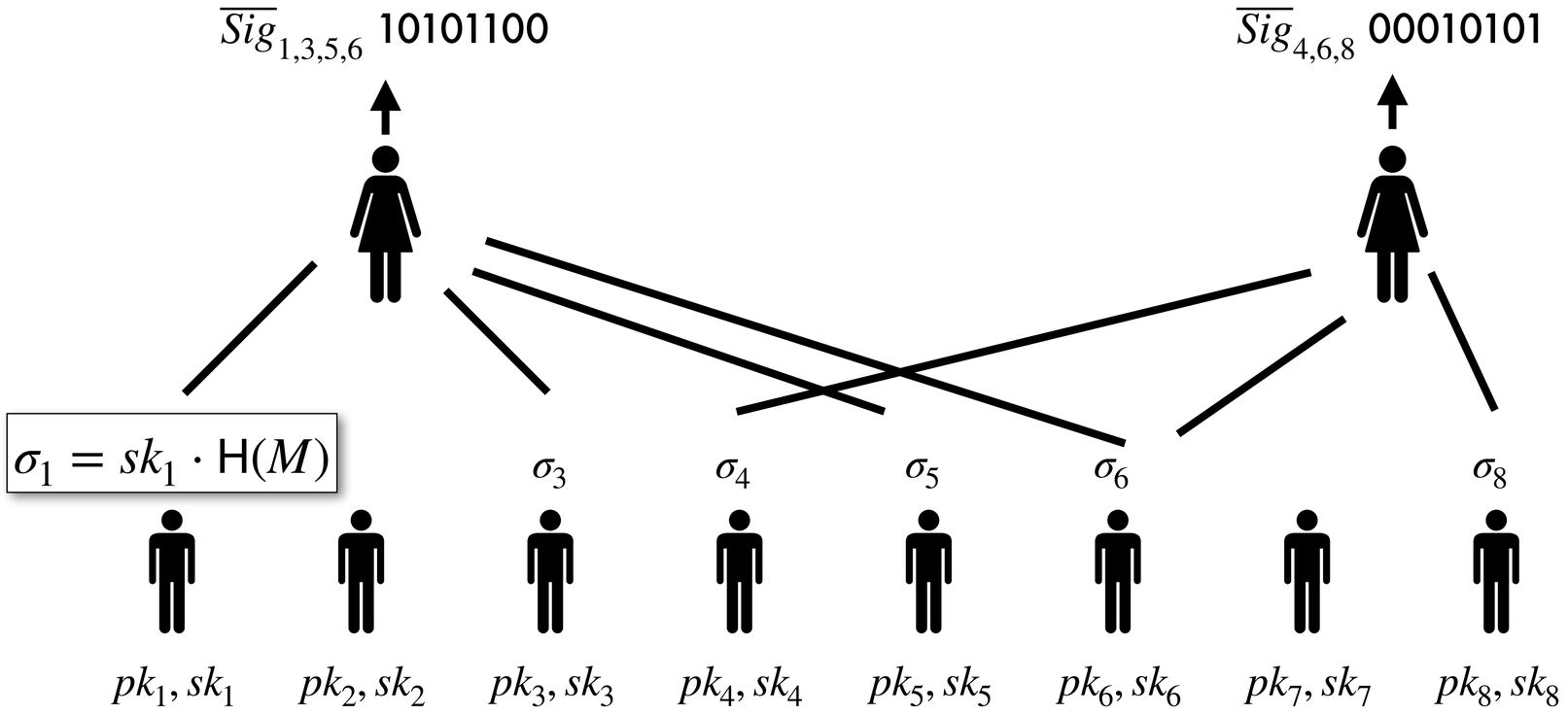
Non-Interactive Multi-Signatures

Example with BLS



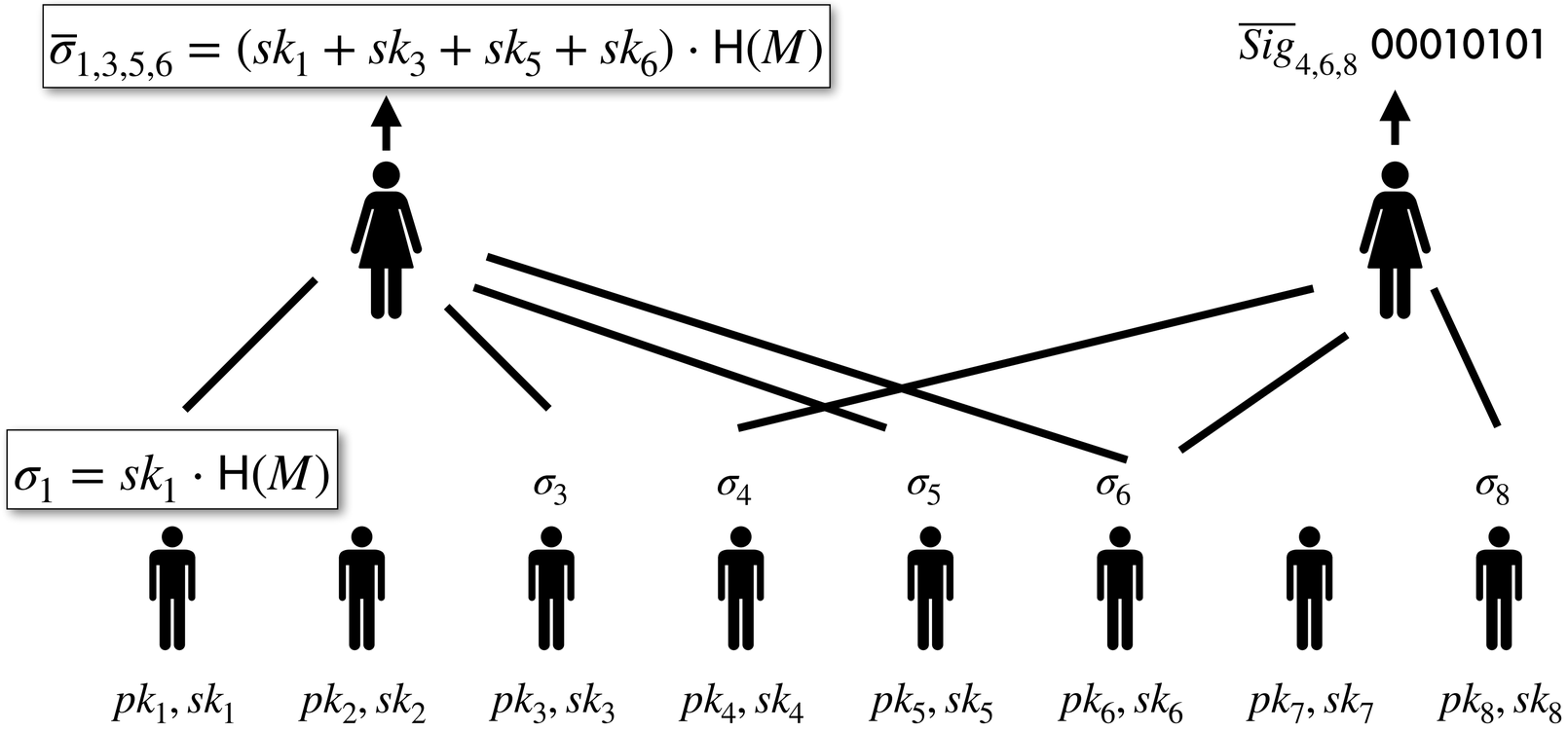
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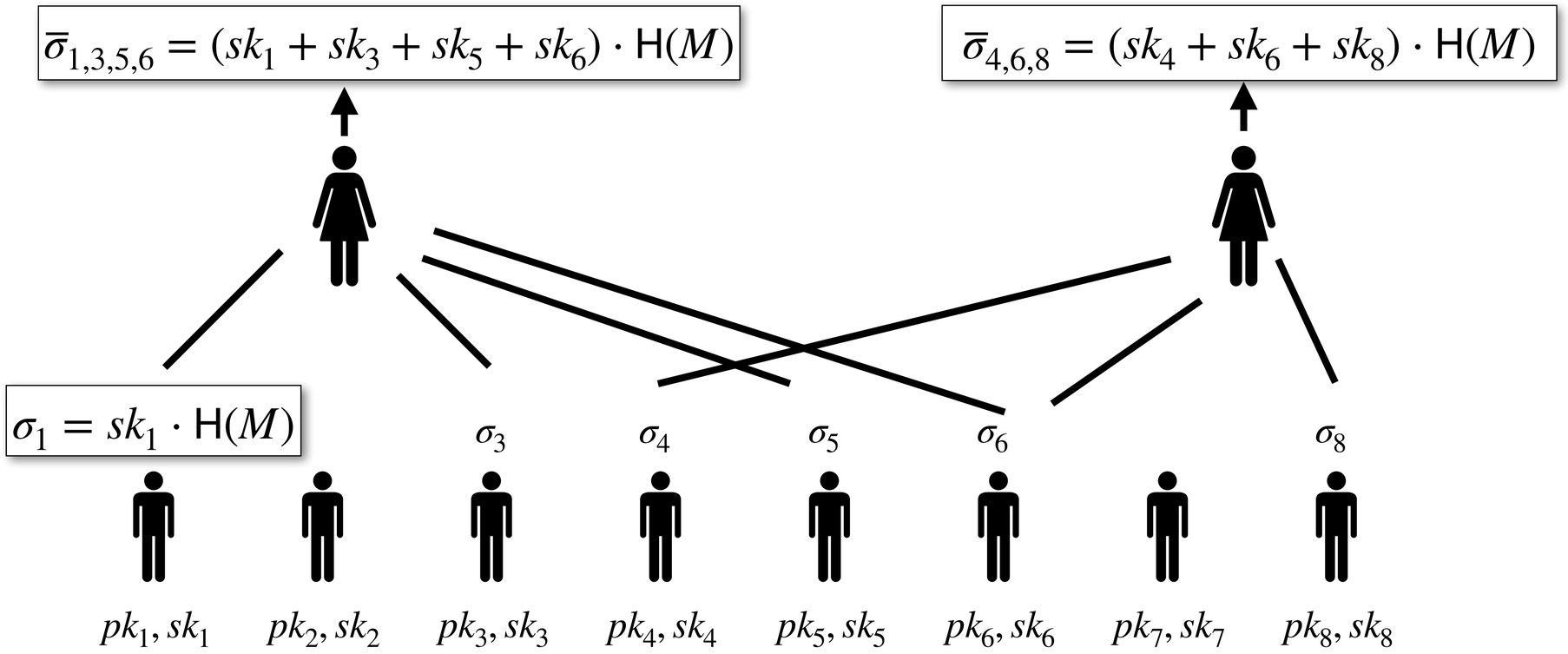
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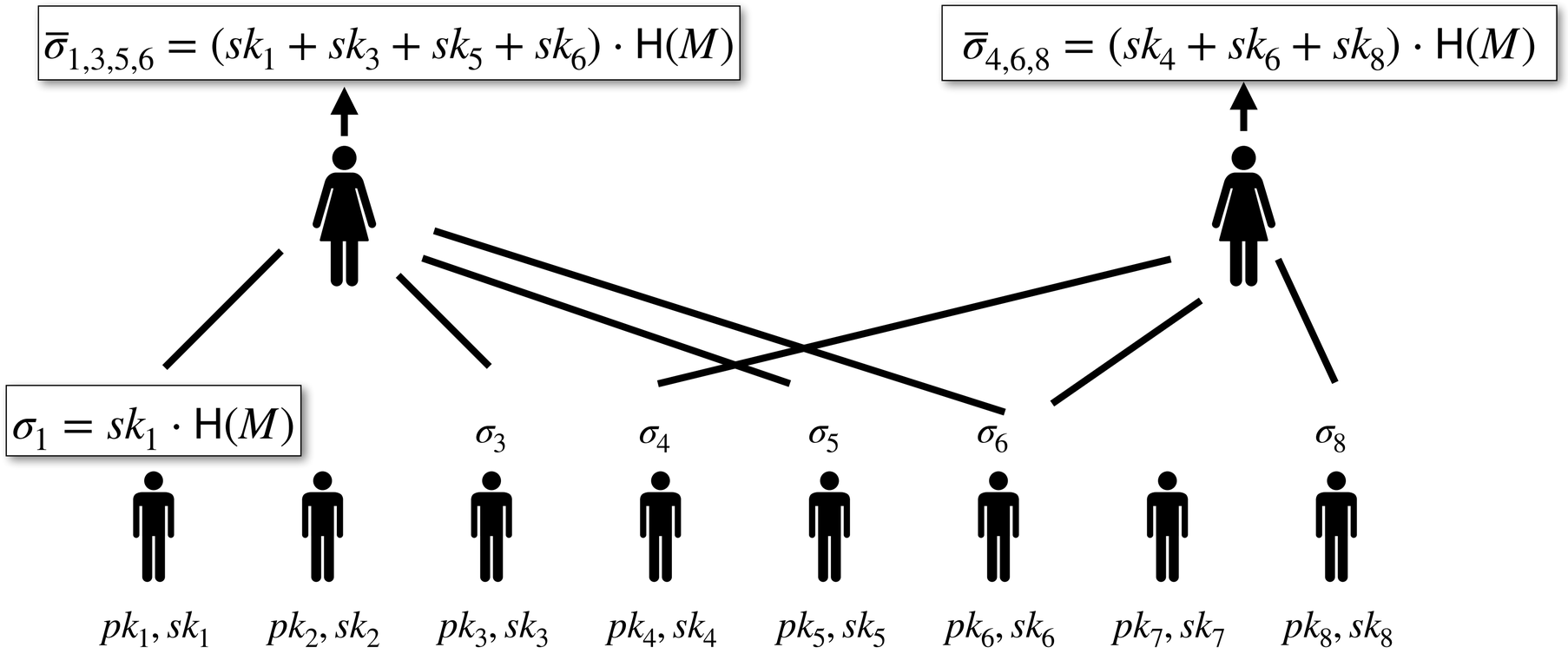
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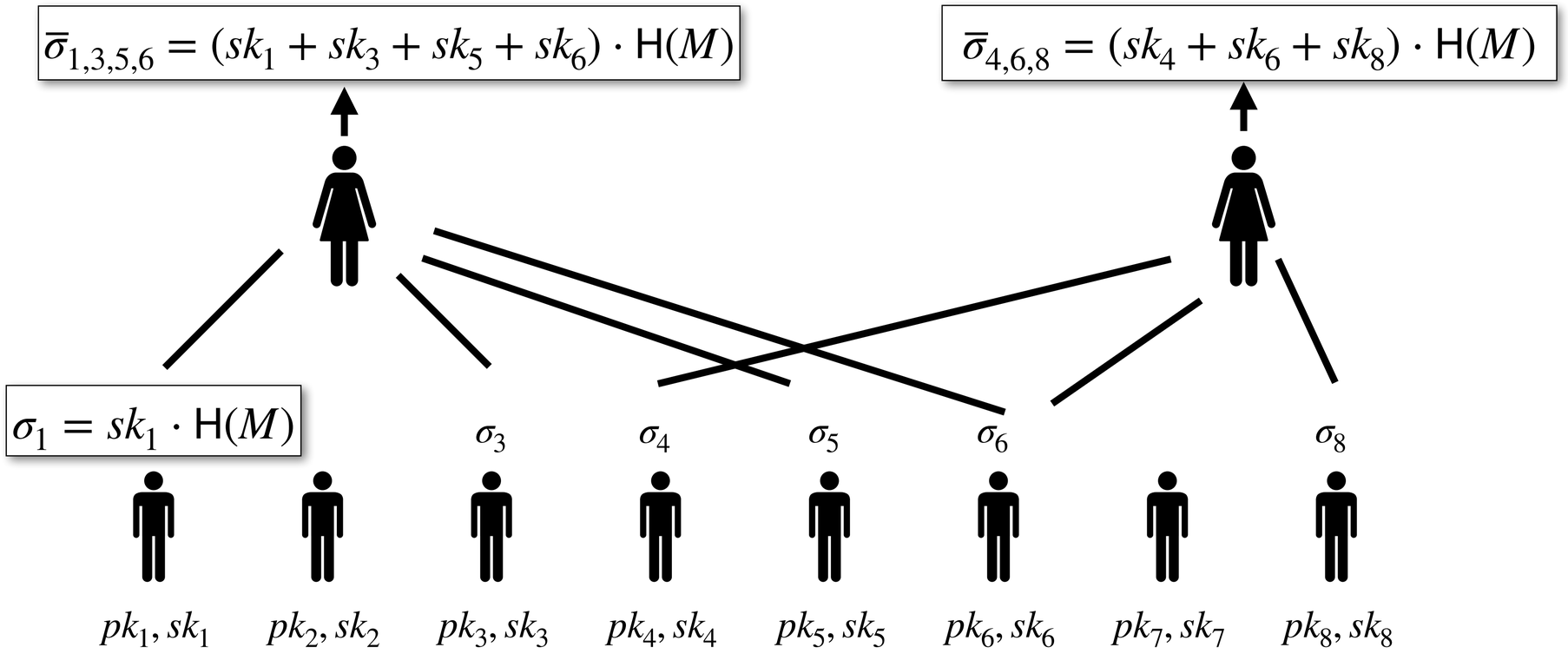
$$\overline{\sigma}_{1,3,5,6} + \overline{\sigma}_{4,6,8} = (sk_1 + sk_3 + sk_4 + sk_5 + 2sk_6 + sk_8) \cdot H(M)$$



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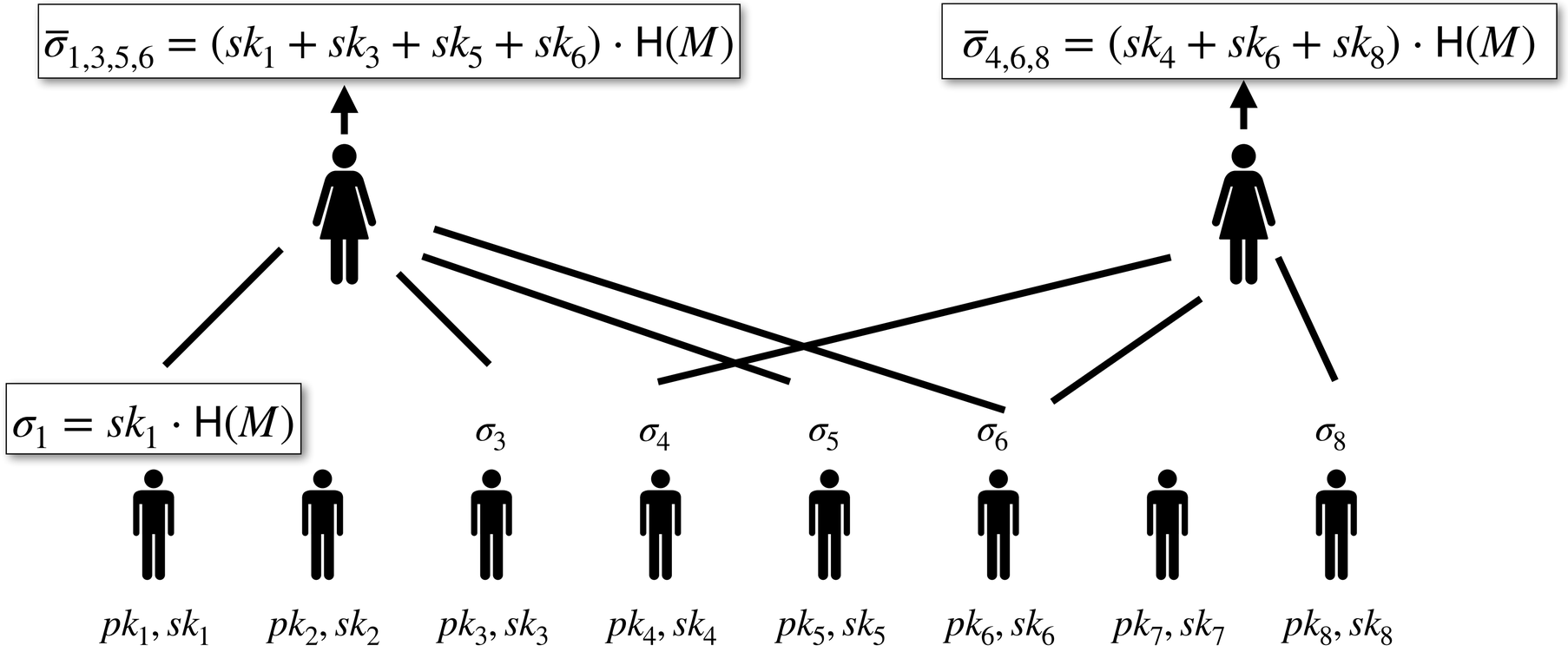
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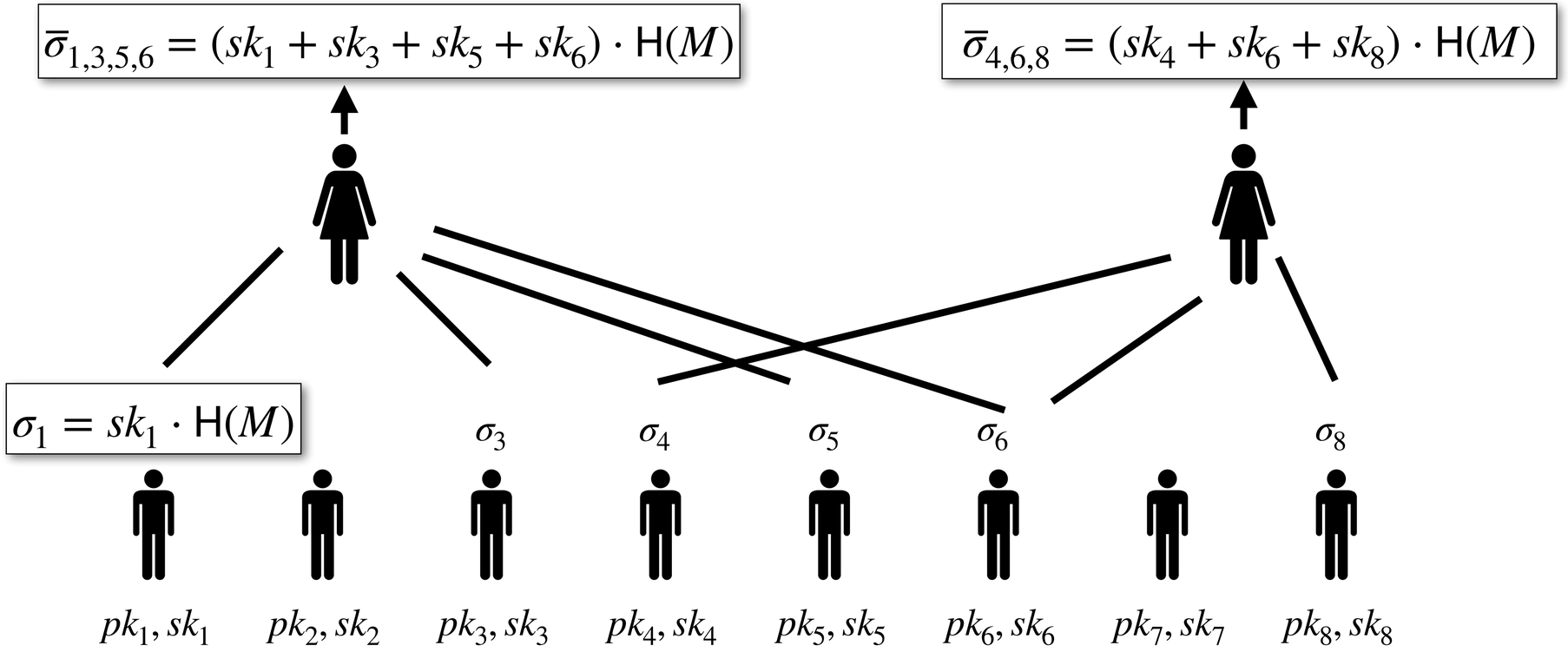


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Counters instead of bitvectors!



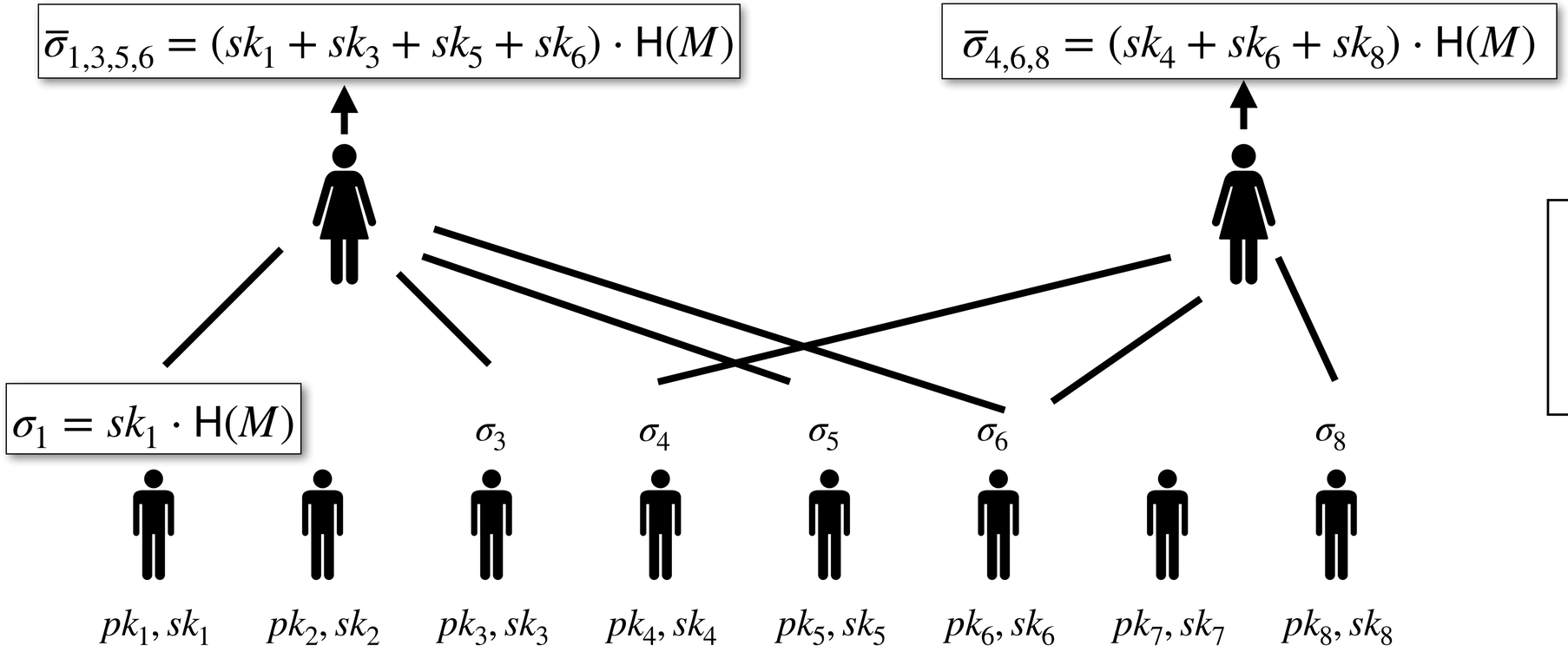
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Scheme with only set of public keys?



This work: Hint-Free Multi-Signatures

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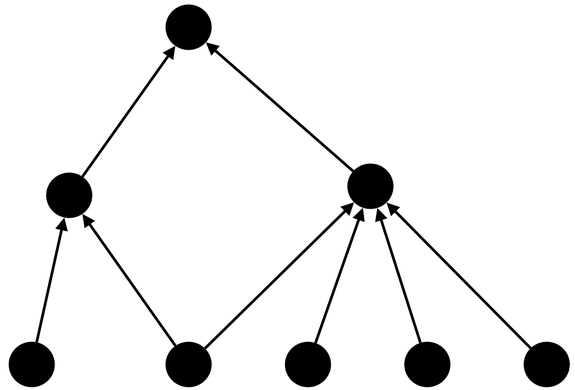
- KeyGen$(1^\lambda) \rightarrow (pk, sk)$

This work: Hint-Free Multi-Signatures

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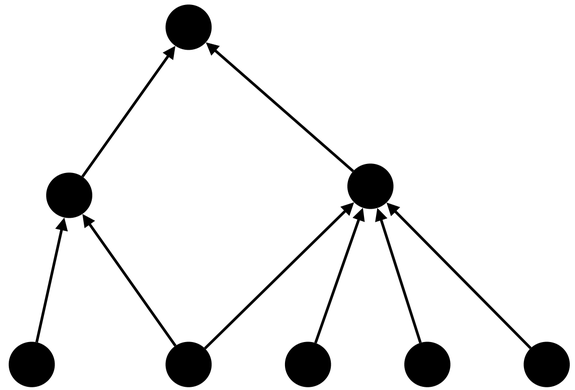
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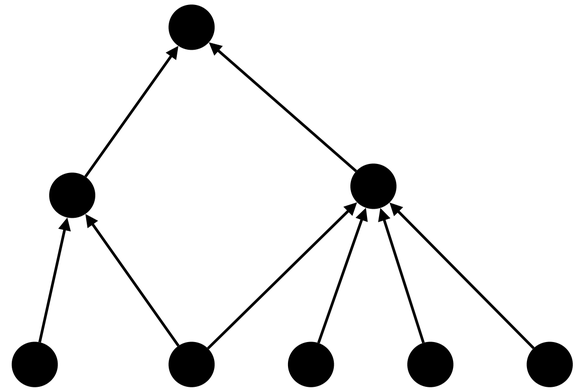
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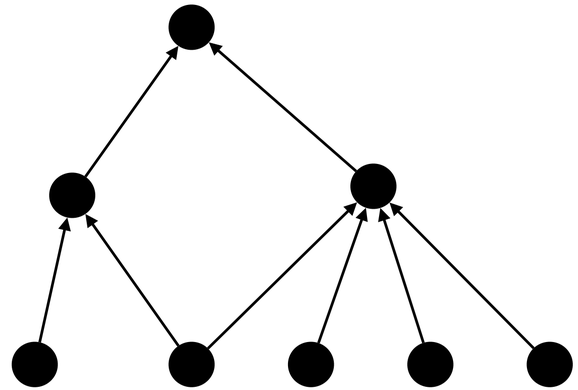
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σ_i may already be aggregated

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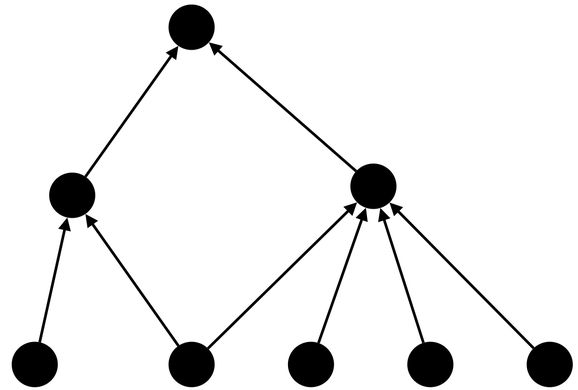
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Keys given as a set
(no multiset, no counters, no topology)

State of the Art

[DKGV22]: Rate-1 Non-Interactive Arguments for Batch-NP and Applications

[this work]: Hint-Free Multi-Signatures

State of the Art

From Pairings / BLS

Need Multiplicities

10111201

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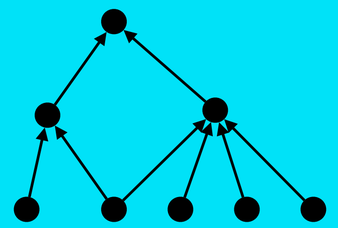
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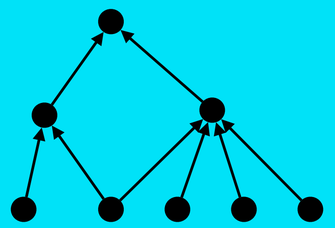
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From Recursive SNARKs

Hint-Free

Signature Agnostic

Security unclear or depth bounded

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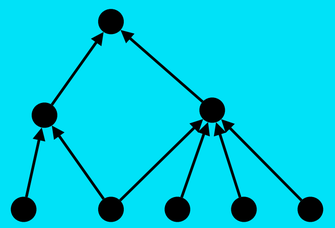
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Hint-Free Multi-Signatures [this work]

Feasibility without Recursion

Not Practical (iO)

[DKGV22]: Rate-1 Non-Interactive Arguments for Batch-NP and Applications

[this work]: Hint-Free Multi-Signatures

Main Construction Idea

Normalization Gadget

Main Construction Idea

Normalization Gadget

Individual

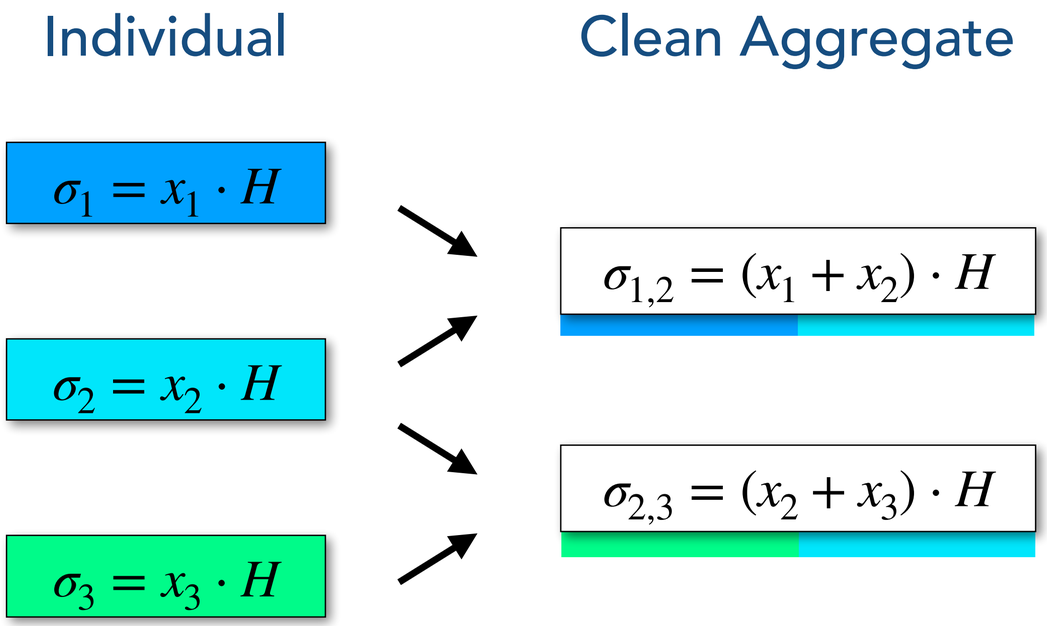
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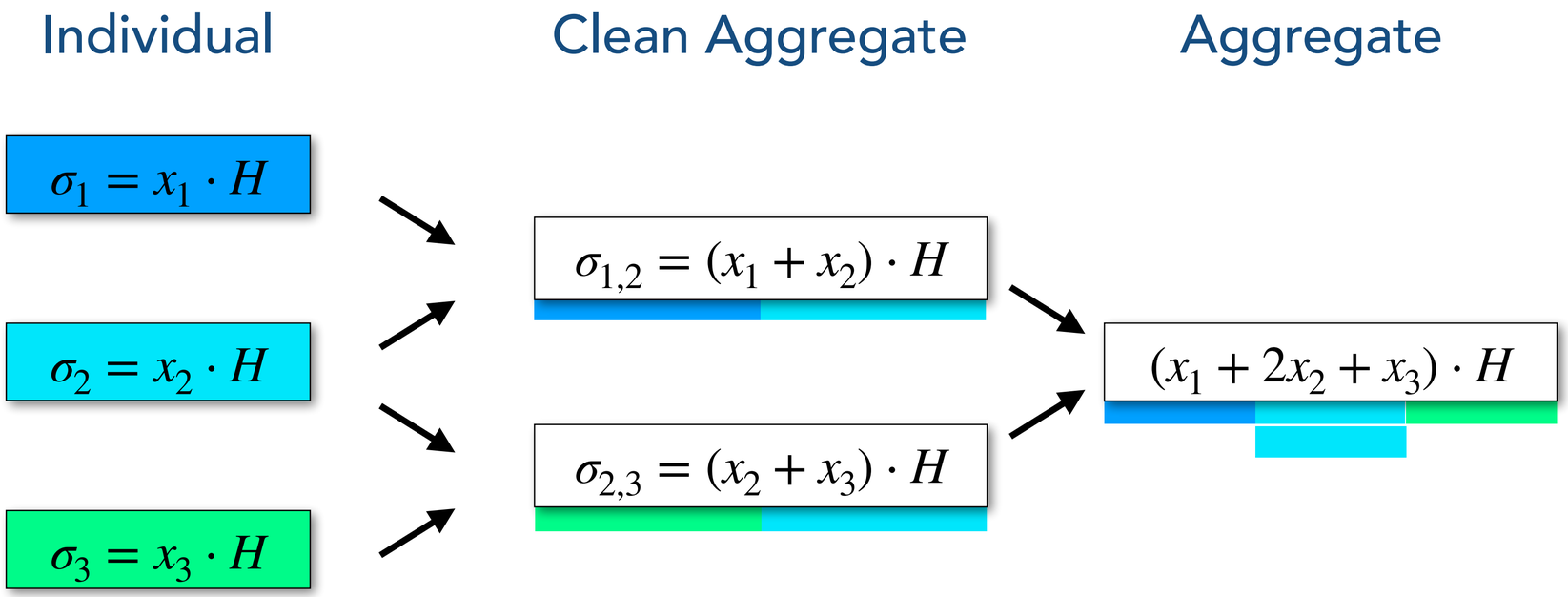
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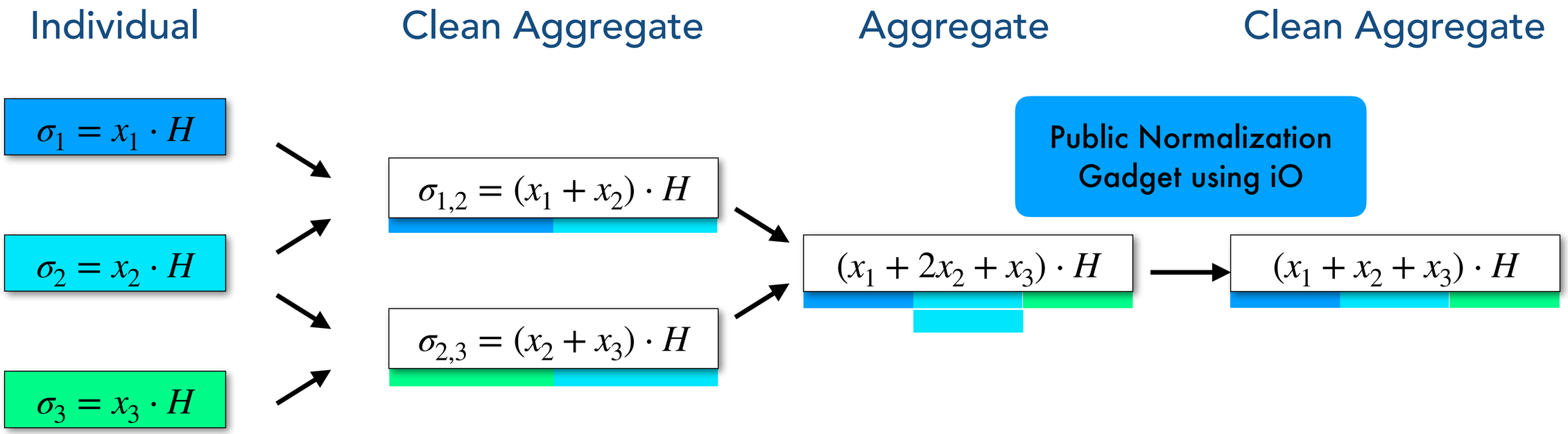
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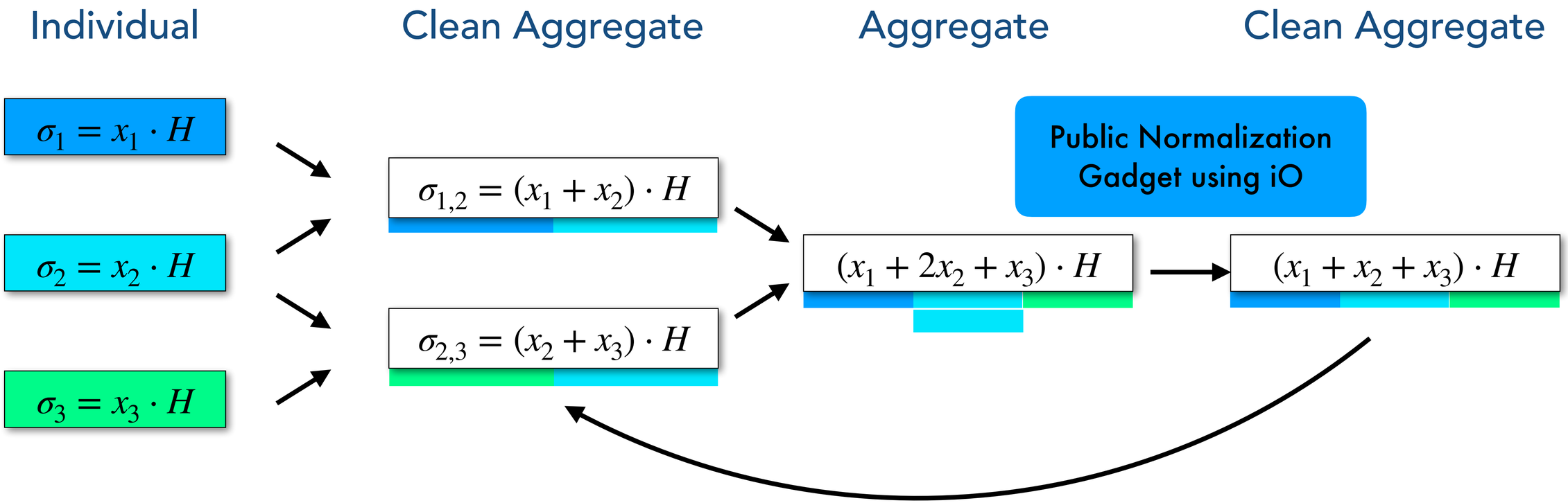
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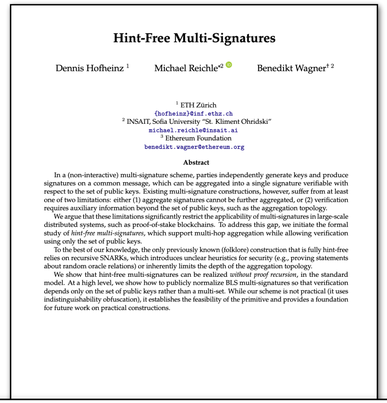


Main Construction Idea

Normalization Gadget



Open Question



Hint-Free Multi-Signatures

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Abstract

In a (non-interactive) multi-signature scheme, parties independently generate keys and produce signatures on a common message, which can be aggregated into a single signature verifiable with respect to the set of public keys. Existing multi-signature constructions, however, suffer from at least one of two limitations: either (1) aggregate signatures cannot be further aggregated, or (2) verification requires auxiliary information beyond the set of public keys, such as the aggregation topology.

We argue that these limitations significantly restrict the applicability of multi-signatures in large-scale distributed systems, such as proof-of-stake blockchains. To address this gap, we initiate the formal study of *hint-free multi-signatures*, which support multi-hop aggregation while allowing verification using only the set of public keys.

To the best of our knowledge, the only previously known (folklore) construction that is fully hint-free relies on recursive SNARKs, which introduces unclear heuristics for security (e.g., proving statements about random oracle relations) or inherently limits the depth of the aggregation topology.

We show that hint-free multi-signatures can be realized *without proof recursion*, in the standard model. At a high level, we show how to publicly normalize BLS multi-signatures so that verification depends only on the set of public keys rather than a multi-set. While our scheme is not practical (it uses indistinguishability obfuscation), it establishes the feasibility of the primitive and provides a foundation for future work on practical constructions.

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Practical hint-free multi-signature scheme?

no recursion, no iO, ideally post-quantum